THE CUBIC AND ITS DISCRIMINANT ON ONE PAGE

Cardano, Δ, and a near-integer from Apéry

One sign decides how many real roots a cubic has, and two cubic-flavoured numbers lead back to almost-integers.

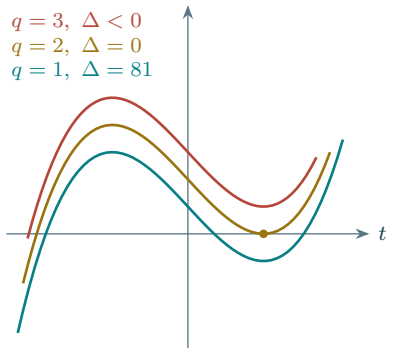
THE RULE

The depressed cubic and Cardano's formula

Every cubic becomes $t^3 + pt + q = 0$ after a shift of x by $\frac{b}{3a}$. Writing $t = u + v$ with $uv = -\frac{p}{3}$ turns it into a quadratic in u^3, whose own discriminant appears under the square root:

$$t = \sqrt[3]{-\frac{q}{2} + \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}} + \sqrt[3]{-\frac{q}{2} - \sqrt{\frac{q^2}{4} + \frac{p^3}{27}}}, \qquad \Delta = -4p^3 - 27q^2$$

Since $\Delta = -108\left(\frac{q^2}{4} + \frac{p^3}{27}\right)$, the sign of Δ is read straight from the square root: $\Delta > 0$ gives three distinct real roots, $\Delta = 0$ a repeated root, and $\Delta < 0$ one real root with two complex partners.



$y = t^3 - 3t + q$: three crossings, a touch at $t = 1$, then a single crossing.

THREE REAL ROOTS

Cosines rescue Cardano

When $\Delta > 0$ the square root in Cardano's formula is imaginary even though all three roots are real, the casus irreducibilis that forced complex numbers into algebra. The identity $4\cos^3\theta - 3\cos\theta = \cos 3\theta$ avoids them: for $t^3 - 3t + 1 = 0$ put $t = 2\cos\theta$, so $\cos 3\theta = -\frac{1}{2}$ and

$$t = 2\cos 40^\circ,\ 2\cos 160^\circ,\ 2\cos 280^\circ = 1.532,\ -1.879,\ 0.347.$$

Here $\Delta = 81 = 9^2$ is a perfect square, the sign that the three roots are rational expressions in one another.

THE GENERAL CASE

The discriminant of $ax^3 + bx^2 + cx + d$

Before depressing the cubic,

$$\Delta = b^2c^2 - 4ac^3 - 4b^3d - 27a^2d^2 + 18abcd,$$

and with roots r_1, r_2, r_3 it equals $a^4 \prod_{i<j}(r_i - r_j)^2$, the squared product of the gaps between roots. That form explains the rule: a repeated root makes a gap zero, and a complex pair makes one squared gap negative. For a quadratic the same recipe gives the familiar $b^2 - 4ac$.

A CUBIC OF DISCRIMINANT -23

The plastic number and a Ramanujan-like identity

The cubic $x^3 = x + 1$ has $p = q = -1$, so $\Delta = 4 - 27 = -23$ and one real root, the plastic number:

$$\rho = \sqrt[3]{\frac{1}{2} + \sqrt{\frac{23}{108}}} + \sqrt[3]{\frac{1}{2} - \sqrt{\frac{23}{108}}} = 1.324\,717\,957\ldots$$

Because $\mathbb{Q}(\sqrt{-23})$ has class number 3 and ρ generates the cubic extension that repairs its factorisation, Weber's modular function satisfies $\mathfrak{f}(\sqrt{-23}) = \sqrt{2}\,\rho$ exactly, and expanding $\mathfrak{f}^{24} = e^{\pi\sqrt{23}} + 24 + \ldots$ gives a relation of the same kind as $e^{\pi\sqrt{163}} \approx 640320^3 + 744$:

$$e^{\pi\sqrt{23}} \quad \approx \quad 2^{12}\rho^{24} \quad - \quad 24, \qquad \text{error } 7.9 \times 10^{-5}$$

A SUGGESTED BUILD FROM APÉRY

Apéry's growth rate makes near-integers

Apéry's numbers $1, 5, 73, 1445, 33001, \ldots$ satisfy $n^3u_n = (34n^3 - 51n^2 + 27n - 5)u_{n-1} - (n-1)^3u_{n-2}$, whose leading terms give $x^2 - 34x + 1 = 0$ and the growth rate

$$\alpha = 17 + 12\sqrt{2} = (1 + \sqrt{2})^4 \approx 33.97.$$

The irrationality proof turns on $\alpha > e^3 \approx 20.09$: Apéry's approximations make $q_n\zeta(3) - p_n$ shrink like α^{-n}, clearing denominators costs about e^{3n}, and so non-zero integer combinations of 1 and $\zeta(3)$ become arbitrarily small, which a rational $\zeta(3)$ would forbid. The conjugate $17 - 12\sqrt{2}$ equals $1/\alpha$, so $\alpha^n + \alpha^{-n}$ is always a whole number and α^n falls short of one by exactly α^{-n}, a Ramanujan-like family with the shortfall known in advance. A modular version would evaluate Beukers' level-6 parametrisation of Apéry's numbers at a CM point, the natural next investigation.

n	$\alpha^n = (17 + 12\sqrt{2})^n$
3	39 201.999 974 5
4	1 331 713.999 999 25
5	45 239 073.999 999 978
6	1 536 796 801.999 999 999 35

Each shortfall is α^{-n}; at $n = 6$ it is 6.5×10^{-10}.

Conventions. $\mathfrak{f}(\tau) = q^{-1/48}\prod_{n\geq 1}(1 + q^{n-1/2})$ with $q = e^{2\pi i\tau}$ is Weber's function, and Apéry's numbers are $u_n = \sum_k \binom{n}{k}^2\binom{n+k}{k}^2$.